MADELYN I. BROOME

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EDUCATION

University of California Santa Cruz, Ph.D. October 2020 - June 2026
Thesis: *Modeling Exoplanet Photoevaporation with X-rays & Metals* - Advised by Ruth Murray-Clay

Cambridge University October 2019 - June 2020
Master's of Advanced Studies in Astrophysics - Graduated with highest mark of Distinction on Thesis (coursework ungraded)
Thesis: *Radiative Transfer in Protoplanetary Disks with Gaps* - Advised by Oliver Shorttle

Princeton University September 2015 - 2019
A.B. Astrophysics - Graduated with Honors
Certificate (minor): Planets and Life
Thesis: *The Destruction of Wide Binaries by the Milky Way* - Advised by David N. Spergel

FELLOWSHIPS AND PRIZES

Total: $373,000

UCSC President's Dissertation-Year Fellowship	2026
Chahta Doctorate & Kanneube Fellowships	2025-26
Eugene Cota-Robles Graduate Fellowship	October 2020-2025
NASA ExoExplorer	2025
ARCS Foundation Fellow	2024
AAS National Osterbrock Leadership Program Fellow	2021-present
UCSC Teaching & Learning Center Graduate Pedagogy Fellow	2023
UCSC Outstanding Physics Community Member	April 2026
UCSC WiSE Denise Denton Women Prize for Excellence in DEI	June 2024
Excellence in Mentoring, UCSC Astro Dept.	2023
Graduate Student Liaison (formerly "Head Grad")	September 2022-2023
Gregory T. Pope Prize for Science Writing	May 2019

PRESENTATIONS (A SELECTION)

Evaporating Worlds: Linking Theory and Observs. of Atmo. Loss - Talk	June 2026
AAS Exoplanet Atmospheres - Dissertation Talk	March 2026
STSci Atmospheric Escape and Replenishment - Talk	October 2025
European Planetary Science Conference - Talk	September 2025
Escape from Exoplanets Conference - Talk	June 2024
AAS Exoplanets V Conference - Talk	June 2024
Early Researchers in Exoplanets - Talk	June 2023
AAS Exoplanets IV Conference - Session talk, poster	June 2022

Posters: Rocky Planets IV (Jan 2026), Extreme Solar Systems III (March 2024), Gordon Conference (June 2023)

Invited: U Michigan Star and Planet Formation seminar (2025), ARCS Annual Meeting (speaker, 2024), International Conference on College Counseling (panelist, 2023), CA STEAM (panelist, 2023)

SELECTION OF PUBLICATIONS

1. **Broome, M.I.**, Shorttle, O., Kama, M., Booth, R.A., "*Iceline Variations Driven by Protoplanetary Disc Gaps*", 2022, MNRAS
2. **Broome, M.I.**, Murray-Clay, R., McCann, J., Owen, J.E., "*A Fast, Open-source 1D Photoevaporation Code with Metal and Multifrequency X-ray Capabilities*", 2026, ApJ
3. **Broome, M.I.**, Murray-Clay, R., Schulick, M., "*Escape of a high-metallicity atmosphere cannot explain HD 189733b's usually deep X-ray transit*", ApJL, available upon request
4. Lloyd, R.O.P., Schreyer, E., Rogers, J., Owen, J.E., **Broome, M.I.**, et al., "*Hydrogen Escaping from a Pair of Exoplanets Smaller than Neptune*", 2025, Nature
5. Tang, Y., Fortney, J., Murray-Clay, R., **Broome, M.I.**, "*Understanding the Origins of Super-Puff Planets: A New Mass-Loss Regime Coupled to Planetary Evolution*", 2025, ApJ
6. Pai Asnodkar A., Wang J., **Broome M.I.**, Huang C., et al., "*PEPSI's non-detection of escaping hydrogen and metal lines adds to the enigma of WASP-12 b*" 2024, MNRAS, 535, 1829.
7. Patra, K.I., Winn, J.N.,..., and **Broome, M.I.**,"*The Continuing Search for Evidence of Tidal Orbital Decay for Hot Jupiters*", ApJ, 2020
8. **Broome, M.I.** & Jue, M., "*Astrobiological Media*", UC Humanities Research Institute Foundry Journal, 2023

PAST RESEARCH

Hunting for Circumnuclear Water Masers with VLA (ALMA, Advisor: V. Impellizerri)	2018
AGN Feedback in a Quasar-driven Superbubble (Advisors: J.E. Greene & A.D. Gould)	2018
Detectability of WASP-12b Orbital Decay w/ TESS (Advisors: J.N. Winn)	2017
Algorithm for Differential Chromatic Refraction (Princeton REU, Advisors: N.B. Lust)	2017

TEACHING & MENTORING

Primary Instructor & **Curriculum Designer:** ASTR 1 - Introduction to the Universe (undergraduate course, 2023), ASTR 205 - Introduction to Teaching and Research (graduate course, 2023, 2024), ASTR 206/Equitable Mentoring for Astro Research Professional Development Fellowship (2024, 2025)

Mentoring: *Research:* Water Worlds Escape (2024-2026), Intro to Astro Research TA (2023), Lamat REU (Summer 2022); *Peer mentor* (2016-2025)

Certifications: *Head TA* (2023-2025), Graduate Pedagogy Fellow, Certificate in Inclusive Pedagogy

OUTREACH (A SMALL SELECTION)

Native Star Stories Night - Lick Observatory - *Program founder*	2023-present
Pyar (public astronomy Python course) - *Instructor*	2020-present
UCSC Ask an Astronomer - *Writer*	2020-present

RESEARCH INTERESTS

Exoplanet Theory (formation and evolution) – Astrostatistics – Geophysics – Astrobiology

ADDITIONAL COMPETENCIES

Languages	Python, C, Fortran, R, MATLAB, IDL, Bash
Software & Tools	HPC (MPI/OpenMP), CASA, GALA, DS9, Github, Rebound, MESA
Software Development	Wind-AE (fully documented, `pip` installable, CI tested, JOSS article)